📊 COMPREHENSIVE DATA DICTIONARY

Water Resources Management - Distrito Federal (DF), Brazil

Purpose: Emergency Planning & Predictive Modeling for Water Reservoir Levels and Rainfall
Date Created: October 19, 2025
Version: 1.0

🎯 EXECUTIVE SUMMARY

This data dictionary catalogs all available data sources for building two predictive models:
1. **Rainfall Prediction Model** - Predict daily/weekly rainfall in DF
2. **Reservoir Level Prediction Model** - Predict water levels in Descoberto and Santa Maria reservoirs

Key Findings:

- **Excellent meteorological data** from INMET (2001, 2015-2024) with hourly observations
- **Rich reservoir telemetry data** from Santa Maria (2014-2021) with 15-minute intervals
- **Comprehensive water supply indicators** from SINISA (2023) for infrastructure and consumption
- **Limited demographic data** from PDAD-DF (2021) for consumption analysis
- **Regulatory context** from historical resolution documents

Critical Data Gaps:

⚠️ **MISSING**: Descoberto reservoir telemetry data (need to extract from Power BI)
⚠️ **MISSING**: Recent INMET data for multiple stations (2019-2024)
⚠️ **MISSING**: Historical water consumption data linked to reservoir operations
⚠️ **MISSING**: Evapotranspiration and soil moisture data

📚 TABLE OF CONTENTS

---

1. DATA SOURCES OVERVIEW

1.1 INMET Meteorological Data (Historical - 2001)

Attribute	Details
Source	Instituto Nacional de Meteorologia (INMET)
File	`2001.zip` → 16 CSV files
Key Station	A001 - BRASILIA
Coverage	2001 (Full year)
Temporal Resolution	Hourly
Records (Brasilia)	8,760 observations
Data Quality	⚠️ Contains -9999 for missing radiation data, comma as decimal separator

Purpose: Establish historical baseline for meteorological patterns in DF region.

1.2 INMET Meteorological Data (Recent - 2015-2024)

Attribute	Details
Source	Instituto Nacional de Meteorologia (INMET)
File	`inmet_A046_GAMA_PONTE_ALTA_2015-2024.csv`
Station	A046 - GAMA PONTE ALTA
Coverage	January 1, 2015 - December 31, 2018 (Note: File name suggests 2015-2024, but data only until 2018)
Temporal Resolution	Hourly
Total Records	87,672 observations
Data Quality	⚠️ Significant missing values (see Section 6)

Purpose: Primary data source for rainfall prediction model training.

1.3 LAI Dam Telemetry Data (Santa Maria)

Attribute	Details
Source	Freedom of Information Act (LAI) Request
File	`Dados das Barragens(1).pdf` (23 MB, 15,500 pages)
Reservoir	Santa Maria (Station 60477100)
Coverage	November 2014 - October 2021
Temporal Resolution	15-minute intervals
Data Quality	✅ High-frequency, consistent measurements
Key Variable	Water level (Cota) in cm and meters

Structure:

```
Station Code | Station Name         | Year | Date       | Time  | Level (cm) | Level (m)
60477100     | SANTA MARIA-BARRAGEM | 2014 | 16/11/2014 | 15:45 | 107084     | 1070.84
```

Purpose: **CRITICAL** for reservoir level prediction model. Contains 7 years of high-frequency water level data.

Note: Pages 10,000+ contain regulatory documents with reservoir capacity percentages and resolutions.

1.4 SINISA Water Supply Data (2023)

Attribute	Details
Source	Sistema Nacional de Informações em Saneamento Básico
File	`SINISA_AGUA_Planilhas_2023_v2.1.1.zip`
Coverage	2023 (Reference year)
Levels	Municipal, State (UF), Federal, Regional providers
Categories	Indicators, Administrative/Financial, Technical Management

Available Data Categories:

1. **Indicators**
   - Water supply coverage (total, urban, rural)
   - Service quality metrics
   - Infrastructure indicators

2. **Administrative & Financial**
   - Operating costs
   - Tariffs and revenues
   - Investment data
   - Employee information

3. **Technical Management**
   - Water production volumes
   - Distribution network characteristics
   - Water losses (physical and commercial)
   - Hydrometering rates
   - Consumption patterns

Purpose: Understand water demand, consumption patterns, and infrastructure capacity.

1.5 SINISA Consolidated Indicators (DF)

Attribute	Details
Source	SINISA
File	`sinisa_indicadores_estruturais_e_operacionais_df.csv`
Records	1 row (DF aggregated data)
Variables	23 operational and structural indicators

Purpose: Quick reference for DF-level water supply performance metrics.

1.6 PDAD-DF Demographic Data (2021)

Attribute	Details
Source	Pesquisa Distrital por Amostra de Domicílios
File	`PDAD-DF_2021.pdf` (3.2 MB)
Coverage	2021
Content	Population, households, socioeconomic data by administrative region

Purpose: Provide demographic context for water consumption analysis.

1.7 DF Report

Attribute	Details
File	`Relatorio_DF.pdf` (2.6 MB)
Content	General information about Distrito Federal

Purpose: Background context on DF characteristics.

1.8 Additional INMET Historical Data

Attribute	Details
Location	`inmet_historico/` folder
Files	ZIP files for years 2015-2024
Status	**Available but not yet extracted**

Purpose: Expand meteorological coverage with multiple stations across DF and surrounding areas.

1.9 Data NOT Yet Available (Critical)

Data Source	Description	Priority	Action Required
Descoberto Reservoir Telemetry	Water level data for Descoberto dam	🔴 CRITICAL	Extract from Power BI dashboard
INMET Recent Multi-Station Data	Extract all stations from 2015-2024 ZIPs	🔴 CRITICAL	Unzip and consolidate
Reservoir Inflow/Outflow	Water flow rates in/out of reservoirs	🟠 HIGH	May be in Power BI or require new LAI request
Water Treatment Plant Data	Daily production volumes by plant	🟠 HIGH	Contact CAESB (water utility)
Evapotranspiration Data	ET rates for water balance calculations	🟡 MEDIUM	Calculate or obtain from agricultural sources

2. COMPLETE VARIABLE CATALOG

2.1 INMET Meteorological Variables

Variable Name (Portuguese)	Variable Name (English)	Unit	Data Type	Description	Importance
DATA (YYYY-MM-DD)	Date	Date	Datetime	Observation date	Key
HORA (UTC)	Hour (UTC)	Hour	String	Observation time in UTC	Key
PRECIPITAÇÃO TOTAL, HORÁRIO (mm)	**Hourly Total Precipitation**	**mm**	**Numeric**	**Total rainfall in the hour**	🔴 **CRITICAL**
PRESSAO ATMOSFERICA AO NIVEL DA ESTACAO, HORARIA (mB)	Station-Level Atmospheric Pressure	mB	Numeric	Atmospheric pressure at station level	High
PRESSÃO ATMOSFERICA MAX.NA HORA ANT. (AUT) (mB)	Max Pressure in Previous Hour	mB	Numeric	Maximum pressure in previous hour	Medium
PRESSÃO ATMOSFERICA MIN. NA HORA ANT. (AUT) (mB)	Min Pressure in Previous Hour	mB	Numeric	Minimum pressure in previous hour	Medium
RADIACAO GLOBAL (KJ/m²)	Global Solar Radiation	kJ/m²	Numeric	Solar radiation received	High
TEMPERATURA DO AR - BULBO SECO, HORARIA (°C)	**Air Temperature (Dry Bulb)**	**°C**	**Numeric**	**Hourly air temperature**	🔴 **CRITICAL**
TEMPERATURA DO PONTO DE		**°C**	**Numeric**	**Temperature at**	🔴 **CRITICAL**

Variable Name (Portuguese)	Variable Name (English)	Unit	Data Type	Description	Importance
ORVALHO (°C)	**Dew Point Temperature**			**which dew forms**	
TEMPERATURA MÁXIMA NA HORA ANT. (AUT) (°C)	Max Temperature in Previous Hour	°C	Numeric	Maximum temp in previous hour	Medium
TEMPERATURA MÍNIMA NA HORA ANT. (AUT) (°C)	Min Temperature in Previous Hour	°C	Numeric	Minimum temp in previous hour	Medium
TEMPERATURA ORVALHO MAX. NA HORA ANT. (AUT) (°C)	Max Dew Point in Previous Hour	°C	Numeric	Max dew point in previous hour	Low
TEMPERATURA ORVALHO MIN. NA HORA ANT. (AUT) (°C)	Min Dew Point in Previous Hour	°C	Numeric	Min dew point in previous hour	Low
UMIDADE REL. MAX. NA HORA ANT. (AUT) (%)	**Max Relative Humidity in Previous Hour**	**%**	**Numeric**	**Maximum RH in previous hour**	🟠 **HIGH**
UMIDADE REL. MIN. NA HORA ANT. (AUT) (%)	**Min Relative Humidity in Previous Hour**	**%**	**Numeric**	**Minimum RH in previous hour**	🟠 **HIGH**
UMIDADE RELATIVA DO AR,		**%**	**Numeric**		🔴 **CRITICAL**

Variable Name (Portuguese)	Variable Name (English)	Unit	Data Type	Description	Importance
HORARIA (%)	**Hourly Relative Humidity**			**Air humidity percentage**	
VENTO, DIREÇÃO HORARIA (gr) (° (gr))	Wind Direction	Degrees	Numeric	Wind direction in degrees	High
VENTO, RAJADA MAXIMA (m/s)	Maximum Wind Gust	m/s	Numeric	Maximum wind gust speed	Medium
VENTO, VELOCIDADE HORARIA (m/s)	Hourly Wind Speed	m/s	Numeric	Average wind speed	High

Note: RADIACAO GLOBAL shows many -9999 values in 2001 data (missing/error code).

2.2 Reservoir Telemetry Variables (LAI Data)

Variable Name	Description	Unit	Data Type	Temporal Resolution	Importance
Código Estação	Station Code	-	String	-	Key
Nome Estação	Station Name	-	String	-	Key
Ano	Year	Year	Integer	-	Key
Data	Date	Date	Date	15 minutes	Key
Hora	Time	Time	Time	15 minutes	Key
Cota (cm)	**Water Level (cm)**	**cm**	**Numeric**	**15 minutes**	🔴 **CRITICAL**
Cota (m)	**Water Level (m)**	**m**	**Numeric**	**15 minutes**	🔴 **CRITICAL**

Available: Santa Maria (60477100) from 2014-2021
Missing: Descoberto reservoir data

2.3 SINISA Water Supply Variables (DF)

2.3.1 Service Coverage Indicators

Code	Variable Name	Unit	Description	Importance
IAG0001	Atendimento da população total com rede de abastecimento de água	%	Total population coverage with water supply network	High
IAG0002	Atendimento da população urbana com rede de abastecimento de água	%	Urban population coverage	High
IAG0003	Atendimento da população rural com rede de abastecimento de água	%	Rural population coverage	Medium
IAG0004	Atendimento dos domicílios totais com rede de abastecimento de água	%	Total household coverage	High
IAG0005	Atendimento dos domicílios urbanos com rede de abastecimento de água	%	Urban household coverage	High
IAG0006	Atendimento dos domicílios rurais com rede de abastecimento de água	%	Rural household coverage	Low

2.3.2 Operational & Structural Indicators (from CSV)

Variable Name	Unit	DF Value (2023)	Importance	Description
Extensão de rede de distribuição de água por ligação	m/connection	12.64	Medium	Distribution network length per connection
Densidade de economias de água por ligação	units/connection	1.55	Low	Water economy density per connection
Incidência da hidrometração de água	%	99.76	High	Water metering coverage rate
Incidência das economias residenciais ativas de água	%	94.64	Medium	Active residential water accounts
Micromedição de água em relação ao volume disponibilizado para distribuição	%	56.10	High	Micrometering vs. available volume
Micromedição do volume de água consumido	%	92.31	High	Micrometered consumption
Macromedição do volume de água de entrada no sistema de distribuição	%	97.96	High	Macrometered input to distribution
Consumo micromedido de água por economia	m³/month/unit	11.34	🟠 HIGH	Metered consumption per unit
Consumo de água faturado por economia	m³/month/unit	12.09	🟠 HIGH	Billed consumption per unit

Variable Name	Unit	DF Value (2023)	Importance	Description
Consumo total médio per capita de água	L/inhabitant/day	186.8	🔴 CRITICAL	Average total water consumption per capita
Consumo residencial médio per capita de água	L/inhabitant/day	139.79	🔴 CRITICAL	Average residential consumption per capita
Volume de água disponibilizado para distribuição por economia	m³/month/unit	20.18	🟠 HIGH	Available volume per unit
Consumo total médio de água por economia	m³/month/unit	12.26	🟠 HIGH	Average consumption per unit
Consumo de água em relação ao volume disponibilizado para distribuição	%	68.54	🔴 CRITICAL	Consumption vs. available volume
Volume faturado de água em relação ao volume de entrada no sistema de distribuição	%	59.91	High	Billed vs. input volume
Perdas de faturamento de água	%	40.09	🔴 CRITICAL	Water billing losses
Perdas totais de água na distribuição	%	31.46	🔴 CRITICAL	Total water losses in distribution
Perdas totais lineares de	m³/km/day	24.03	🟠 HIGH	Linear water losses

Variable Name	Unit	DF Value (2023)	Importance	Description
água na rede de distribuição				
Perdas totais de água por ligação	L/connection/day	324.39	High	Water losses per connection
Incidência de ligações de água setorizadas	%	31.85	Medium	Sectorized connection rate
Consumo médio de energia elétrica no serviço de abastecimento de água	kWh/m³	0.87	Medium	Energy consumption in water supply

Key Insights:
- DF has high metering coverage (99.76%)
- Significant water losses: 31.46% total, 40.09% billing losses
- High per capita consumption: 186.8 L/day (above national average)
- Water consumption is 68.54% of available volume

---

3. PRIORITY VARIABLES FOR RAINFALL PREDICTION

🎯 Ranking by Importance

Tier 1: Essential (Core Features) 🔴

Rank	Variable	Source	Justification	Availability
1	**Historical Precipitation**	INMET	Direct target variable; temporal patterns, seasonality	✅ Available
2	**Relative Humidity**	INMET	Strong correlation with rainfall probability; moisture availability	✅ Available
3	**Dew Point Temperature**	INMET	Indicates atmospheric moisture; rainfall formation potential	✅ Available
4	**Air Temperature**	INMET	Affects convection and precipitation patterns	✅ Available
5	**Atmospheric Pressure**	INMET	Pressure systems drive weather patterns	✅ Available

Derived Features (Essential):
- **Humidity Trend** (change over 3, 6, 12, 24 hours)
- **Temperature Gradient** (change rate)
- **Pressure Tendency** (rising/falling/steady)
- **Cumulative Rainfall** (last 7, 14, 30 days)
- **Dry Days Count** (consecutive days without rain)
- **Temporal Features**: Day of year, Month, Season (wet/dry)

Tier 2: Important (Enhancement Features) 🟠

Rank	Variable	Source	Justification	Availability
6	**Wind Speed & Direction**	INMET	Transport of moisture systems; regional patterns	✅ Available
7	**Solar Radiation**	INMET	Energy source for convection; evapotranspiration	⚠️ Partial (missing 2001)
8	**Temperature Range**	INMET	Daily variation indicates atmospheric stability	✅ Available (derived)
9	**Wet Bulb Temperature**	INMET	Can be calculated from temp + humidity	✅ Derivable
10	**Precipitation Intensity Patterns**	INMET	Heavy vs. light rain classification	✅ Derivable

Derived Features (Important):
- **Moisture Convergence Index** (humidity × wind patterns)
- **Diurnal Temperature Range** (DTR)
- **Evapotranspiration Estimate** (from temp, humidity, radiation)
- **Atmospheric Instability Index** (from temp and humidity profiles)

Tier 3: Beneficial (Contextual Features) 🟡

Rank	Variable	Source	Justification	Availability
11	**Reservoir Levels**	LAI/Power BI	Indirect indicator of regional precipitation	⚠️ Partial
12	**El Niño/La Niña Index**	External	Large-scale climate patterns affecting Brazil	🔍 Need to obtain
13	**Soil Moisture**	External	Affects local evapotranspiration	❌ Not available
14	**Vegetation Index (NDVI)**	Satellite	Proxy for ET and surface conditions	🔍 Could obtain
15	**Regional Precipitation**	INMET Multi-station	Spatial patterns and system movement	⚠️ Need to process

---

📊 Feature Engineering Recommendations

Temporal Aggregations

```
# Rolling statistics
- Rainfall: sum, mean, max (3h, 6h, 12h, 24h, 7d, 30d windows)
- Temperature: mean, min, max, std (6h, 24h windows)
- Humidity: mean, min, max (3h, 6h, 24h windows)
- Pressure: mean, trend (6h, 12h, 24h windows)

# Lag features
- Precipitation: t-1h, t-3h, t-6h, t-12h, t-24h, t-48h
- Humidity: t-1h, t-3h, t-6h
- Temperature: t-3h, t-6h, t-12h
- Pressure: t-3h, t-6h, t-12h
```

Cyclical Encoding

```
# For temporal features
- Hour of day: sin/cos transformation (0-23)
- Day of year: sin/cos transformation (1-365)
- Month: sin/cos transformation (1-12)
```

Interaction Features

```
- Temperature × Humidity (heat index proxy)
- Pressure Change × Humidity (instability indicator)
- Wind Speed × Humidity (moisture transport)
- Temperature Range × Humidity Range (atmospheric dynamics)
```

---

4. PRIORITY VARIABLES FOR RESERVOIR LEVEL PREDICTION

🎯 Ranking by Importance

Tier 1: Essential (Core Features) 🔴

Rank	Variable	Source	Justification	Availability
1	**Historical Reservoir Level**	LAI/Power BI	Direct target; autoregressive patterns	⚠️ Partial (Santa Maria only)
2	**Precipitation in Watershed**	INMET	Primary input to reservoir (runoff)	✅ Available
3	**Water Consumption**	SINISA	Primary outflow from reservoir	⚠️ Annual only (need daily)
4	**Cumulative Rainfall**	INMET	Watershed saturation and runoff coefficient	✅ Derivable
5	**Days Since Last Rain**	INMET	Affects runoff response	✅ Derivable

Derived Features (Essential):
- **Reservoir Level Trend** (change rate: hourly, daily, weekly)
- **Reservoir Fill Rate** (level change per mm of rain)
- **Estimated Daily Water Demand** (from consumption patterns)
- **Seasonal Component** (reservoir follows annual cycle)
- **Reservoir Drawdown Rate** (during dry periods)

Tier 2: Important (Enhancement Features) 🟠

Rank	Variable	Source	Justification	Availability
6	**Evaporation Rate**	Derived/External	Water loss from reservoir surface	🔍 Need to calculate/obtain
7	**Temperature**	INMET	Affects evaporation and water demand	✅ Available
8	**Relative Humidity**	INMET	Affects evaporation rate	✅ Available
9	**Wind Speed**	INMET	Increases evaporation from surface	✅ Available
10	**Water Treatment Plant Production**	External	Actual withdrawal from reservoir	🔍 Need to obtain
11	**Population Served**	PDAD-DF	Drives water demand	✅ Available (static)
12	**Solar Radiation**	INMET	Drives evaporation	⚠️ Partial

Derived Features (Important):
- **Watershed Wetness Index** (antecedent moisture)
- **Pan Evaporation Estimate** (from temp, humidity, wind, radiation)
- **Seasonal Water Demand Pattern** (from SINISA annual data)
- **Reservoir Capacity Percentage** (level/max capacity)
- **Inter-Reservoir Correlation** (Santa Maria vs. Descoberto patterns)

Tier 3: Beneficial (Contextual Features) 🟡

Rank	Variable	Source	Justification	Availability
13	**Upstream Flow/Inflow**	External	Direct measurement of water input	🔍 Need to obtain
14	**Water Losses in Distribution**	SINISA	Reduces actual consumption from reservoir	✅ Available (31.46%)
15	**Socioeconomic Factors**	PDAD-DF	Affect consumption patterns	✅ Available
16	**Watershed Land Use**	Satellite	Affects runoff coefficient	🔍 Could obtain
17	**Soil Saturation**	External	Affects runoff response	❌ Not available
18	**Regulatory Restrictions**	Historical docs	Water rationing events	⚠️ In PDF (pages 10000+)

📊 Feature Engineering Recommendations

Reservoir-Specific Features

```
# Level dynamics
- Level change: t-1d, t-7d, t-30d
- Level velocity: (level_t - level_t-1) / time_diff
- Level acceleration: velocity_t - velocity_t-1
- Fill/drain rate classification: filling, stable, draining

# Capacity features
- Current capacity percentage
- Distance from critical levels (alert, attention, emergency)
- Days until empty (at current drain rate)
- Days until full (at current fill rate)
```

Rainfall-Runoff Features

```
# Antecedent moisture
- API (Antecedent Precipitation Index): weighted sum of past rainfall
- Soil moisture proxy: exponential decay of rainfall
- Dry days count: consecutive days with rain < 1mm

# Watershed response
- Weighted rainfall: closer stations weighted higher
- Rainfall intensity: hourly peak vs. daily average
- Storm characteristics: duration, intensity, total volume
```

Demand Features

```
# Temporal patterns
- Day of week: weekday vs. weekend demand
- Month: seasonal consumption variation
- Holiday indicator: reduced consumption
- Growing season: increased demand (if applicable)

# Consumption estimation
- Per capita demand × population
- Adjust for water losses (31.46%)
- Temperature-adjusted demand (higher in hot/dry periods)
```

Evaporation Features

```
# Penman-Monteith or simplified equation
- Inputs: Temperature, Humidity, Wind Speed, Solar Radiation
- Pan evaporation equivalent
- Reservoir surface area factor
- Daily/monthly evaporation estimate
```

---

5. RELATIONSHIPS & DEPENDENCIES

5.1 System Conceptual Model

```
┌─────────────────────────────────────────────────────────────────┐
│                    RAINFALL (INMET Data)                         │
│    • Precipitation intensity & duration                          │
│    • Spatial distribution across watershed                       │
└─────────────────┬───────────────────────────────────────────────┘
                  │
                  ↓
┌─────────────────────────────────────────────────────────────────┐
│                 WATERSHED CHARACTERISTICS                        │
│    • Runoff coefficient (depends on soil saturation)             │
│    • Time of concentration                                       │
│    • Land use / vegetation                                       │
└─────────────────┬───────────────────────────────────────────────┘
                  │
                  ↓
┌─────────────────────────────────────────────────────────────────┐
│           RESERVOIR INFLOW (Not directly measured)               │
│    • Surface runoff from rainfall                                │
│    • Baseflow from groundwater                                   │
│    • Direct precipitation on reservoir surface                   │
└─────────────────┬───────────────────────────────────────────────┘
                  │
                  ↓
┌─────────────────────────────────────────────────────────────────┐
│         RESERVOIR STORAGE (LAI Telemetry Data)                   │
│    Santa Maria: ✅ 2014-2021 (15-min intervals)                  │
│    Descoberto:  ❌ Need to extract from Power BI                 │
│                                                                  │
│    Level = Level_t-1 + Inflow - Outflow - Evaporation            │
└─────────────────┬───────────────────────────────────────────────┘
                  │
                  ↓
┌─────────────────────────────────────────────────────────────────┐
│           WATER TREATMENT & DISTRIBUTION                         │
│    • Production by treatment plants                              │
│    • Distribution to population (SINISA: 186.8 L/cap/day)        │
│    • Network losses (SINISA: 31.46%)                             │
└─────────────────┬───────────────────────────────────────────────┘
                  │
                  ↓
┌─────────────────────────────────────────────────────────────────┐
│                 END-USE CONSUMPTION                              │
│    • Residential (SINISA: 139.79 L/cap/day)                      │
│    • Commercial                                                  │
│    • Industrial                                                  │
│    • Public use                                                  │
└─────────────────────────────────────────────────────────────────┘

PARALLEL OUTFLOWS FROM RESERVOIR:
┌─────────────────────────────────────────────────────────────────┐
│                   EVAPORATION LOSSES                             │
│    • Temperature-dependent (INMET: Temperature, Humidity)        │
│    • Wind-enhanced (INMET: Wind Speed)                           │
│    • Radiation-driven (INMET: Solar Radiation)                   │
│    • Surface area-dependent (Reservoir level → surface area)     │
└─────────────────────────────────────────────────────────────────┘
```

5.2 Key Relationships

5.2.1 Rainfall → Reservoir Level

Time Lag:
- Immediate (direct rainfall on reservoir): 0-3 hours
- Surface runoff: 6-24 hours (depends on distance from reservoir)
- Baseflow: days to weeks

Relationship Strength:

```
Reservoir_Level_Change ∝ (Rainfall - Threshold) × Runoff_Coefficient × Watershed_Area
```

Non-Linearity:
- Initial rainfall (first 5-10mm) absorbed by soil → low runoff
- Soil saturation → runoff coefficient increases dramatically
- Intensity matters: 20mm in 1 hour ≠ 20mm in 24 hours

Spatial Consideration:
- Rainfall must be weighted by proximity to reservoir
- Upstream precipitation more important than downstream
- Need to aggregate multiple INMET stations

5.2.2 Reservoir Level → Water Consumption

Direct Relationship:

```
Daily_Withdrawal = Population × Per_Capita_Consumption / (1 - Loss_Rate)
                 = 3,000,000 × 186.8 L/day / (1 - 0.3146)
                 ≈ 817 million liters/day
                 ≈ 817,000 m³/day
```

Modifying Factors:
- **Temperature**: +1°C → ~2-5% increase in consumption
- **Day of Week**: Weekend consumption ~10-15% lower
- **Season**: Dry season (May-Sept) → higher consumption
- **Restrictions**: Regulatory limits during crises

Emergency Thresholds (from regulatory docs in LAI PDF):
- **100%**: Full capacity - normal operation
- **<60-70%**: Attention level - monitoring increased
- **<30-40%**: Alert level - restrictions may begin
- **<20-30%**: Emergency level - rationing enacted

5.2.3 Rainfall → Water Consumption (Indirect)

Relationship:
- **Rainfall ↑ → Outdoor water use ↓** (no need for gardening/car washing)

- **Dry period ↑ → Consumption ↑** (irrigation, cooling)
- **Extreme heat + no rain → Peak consumption** (can be +30-50% above baseline)

5.2.4 Water Consumption → Costs/Expenses

SINISA Financial Data (Annual, need to extract from Excel sheets):
- Operating costs per m^3
- Energy costs (0.87 kWh/m^3 × electricity tariff)
- Treatment costs
- Maintenance costs
- Revenue from tariffs

Relationship:

```
Total_Cost = Fixed_Costs + (Variable_Cost_per_m³ × Volume_Produced)
           + (Energy_Cost_per_kWh × 0.87 kWh/m³ × Volume)
           + Treatment_Chemicals
           + Maintenance
```

Emergency Costs:
- Low reservoir levels → increased pumping costs (lower water = higher lift)
- Crisis response costs (tanker trucks, temporary infrastructure)

5.3 Feedback Loops

Loop 1: Reservoir Crisis Response

```
Low Reservoir Level
    ↓
Restrictions Announced
    ↓
Consumption Decreases
    ↓
Reservoir Drawdown Slows
    ↓
Rainfall has larger relative impact
    ↓
Recovery accelerated
```

Loop 2: Dry Season Amplification

```
No Rainfall
    ↓
Soil dries
    ↓
Temperature increases
    ↓
Evaporation increases
    ↓
Reservoir level decreases faster
    ↓
Consumption may increase (heat)
    ↓
Accelerated depletion
```

---

6. DATA QUALITY ASSESSMENT

6.1 INMET Data Quality Issues

Missing Values Analysis (2015-2018 dataset, 87,672 records)

Variable	Missing Count	Missing %	Severity
DATE & TIME columns	52,608	60.0%	🟡 MEDIUM (appears to be duplicate columns)
RADIACAO GLOBAL (KJ/m²)	64,476	73.5%	🔴 HIGH
TEMPERATURA DO AR	953	1.1%	🟢 LOW
TEMPERATURA PONTO ORVALHO	953	1.1%	🟢 LOW
PRECIPITAÇÃO	1,286	1.5%	🟢 LOW
PRESSAO ATMOSFER-ICA	2,393	2.7%	🟢 LOW
UMIDADE RELATIVA	953	1.1%	🟢 LOW
VENTO VELOCIDADE	992	1.1%	🟢 LOW
VENTO DIREÇÃO	1,018	1.2%	🟢 LOW

Issues Identified:

1. **Solar Radiation**:
   - 73.5% missing in 2015-2018 data

- -9999 sentinel values in 2001 data
- **Impact**: Cannot use for evaporation calculations without imputation
- **Mitigation**: Use alternative ET calculation methods or interpolate from nearby stations

2. **Duplicate/Redundant Columns**:
   - Several "Data" and "Hora" columns with different names
   - 60% missing suggests they are not the primary date columns
   - **Action**: Use "DATA (YYYY-MM-DD)" and "HORA (UTC)" as primary keys

3. **Numeric Data Format Issues**:
   - **Comma as decimal separator** (e.g., "1070,84" instead of "1070.84")
   - **Mixed text and numbers** (e.g., ",3" instead of "0.3")
   - **Impact**: Data type errors, need string cleaning before conversion
   - **Mitigation**: Replace commas, handle leading/trailing commas

4. **Temporal Coverage Gap**:
   - File named "2015-2024" but only contains data until 2018
   - **Missing**: 2019-2024 data
   - **Action**: Extract from `inmet_historico/` ZIP files (2019-2024)

6.2 LAI Telemetry Data Quality

Strengths:
✅ High-frequency measurements (15-minute intervals)
✅ Consistent format throughout 15,500 pages
✅ Long time series (2014-2021 = 7 years)
✅ Appears complete for Santa Maria

Weaknesses:
❌ **Only Santa Maria reservoir covered** - Descoberto missing
❌ **PDF format** - requires extraction to structured format
❌ **Pages 10,000+** contain unrelated regulatory documents
❌ **No metadata** - need to obtain reservoir capacity, surface area curves, operating rules

Required Actions:
1. Extract all Santa Maria telemetry to CSV (pages 1-10,000)
2. Obtain Descoberto telemetry from Power BI or new LAI request
3. Get reservoir specifications:
- Maximum capacity (m³)
- Surface area vs. level curve
- Dead volume level
- Operating rules and target levels

6.3 SINISA Data Quality

Strengths:
✅ Comprehensive indicators for water supply system
✅ Official government data source
✅ DF-specific aggregation available
✅ Detailed technical documentation

Weaknesses:
❌ **Annual temporal resolution only** (not suitable for daily prediction)
❌ **2023 reference year only** - no time series
❌ **Excel format with complex structure** (multiple header rows, merged cells)
❌ **Aggregated at state level** - no facility-level detail
❌ **No direct link to reservoir operations**

Required Actions:
1. Extract DF-specific values from all sheets
2. Use as static context (consumption rates, loss percentages)
3. Obtain daily/monthly consumption data from CAESB (water utility)
4. Link consumption patterns to reservoir withdrawals

6.4 Demographic Data Quality (PDAD-DF)

Strengths:
✅ Detailed demographic breakdown by administrative region
✅ Comprehensive socioeconomic indicators
✅ Official government survey

Weaknesses:
❌ **2021 reference year only** - snapshot, not time series
❌ **PDF format** - requires manual extraction
❌ **Not directly linked to water consumption data**

Use Case:
- Population figures for per capita calculations
- Socioeconomic context for consumption patterns
- Regional distribution for spatial analysis

6.5 Data Gaps Summary

Data Need	Criticality	Current Status	Recommended Action
Descoberto Reservoir Levels	🔴 CRITICAL	Missing	Extract from Power BI dashboard (already collected)
INMET Multi-Station Data (2019-2024)	🔴 CRITICAL	Available but not processed	Extract from `inmet_historico/` ZIPs
Daily Water Consumption	🔴 CRITICAL	Only annual aggregate	Request from CAESB or infer from reservoir drawdown
Reservoir Specifications	🟠 HIGH	Missing	Search technical docs or contact dam operator (ADASA)
Evaporation Data	🟠 HIGH	Missing	Calculate using Penman-Monteith with INMET data
Inflow/Outflow Measurements	🟠 HIGH	Missing	May exist at ADASA (dam regulator)
Soil Moisture	🟡 MEDIUM	Missing	Optional - use rainfall proxies
Land Use / NDVI	🟡 MEDIUM	Missing	Optional - satellite data available

---

7. RECOMMENDATIONS

7.1 Immediate Actions (Priority 1) 🔴

1. Extract Descoberto Reservoir Data

Why: Critical for reservoir prediction model; represents ~50% of DF water supply
Action: Use browser automation to extract time series from Power BI dashboard
Timeline: Week 1

2. Process INMET Historical Data (2019-2024)

Why: Need recent data for model training; close the 2019-2024 gap
Action:
- Unzip all files in `inmet_historico/` folder
- Consolidate into single dataset with multiple stations

- Focus on stations closest to reservoirs: Brasilia (A001), Gama (A046), and nearby
Timeline: Week 1

3. Extract Santa Maria Telemetry from PDF

Why: Convert 15,500-page PDF to usable CSV format
Action:
- Python script to parse PDF pages 1-10,000
- Validate data integrity (check for parsing errors)
- Convert to structured CSV with proper datetime
Timeline: Week 1

4. Clean and Consolidate INMET Data

Why: Address data quality issues (commas, missing values)
Action:
- Standardize decimal notation (replace commas)
- Handle missing radiation data (impute or exclude)
- Create unified dataset with consistent timestamps (UTC → local time)
- Quality control checks (outliers, impossible values)
Timeline: Week 1-2

7.2 Short-Term Actions (Priority 2) 🟠

5. Obtain Reservoir Technical Specifications

Why: Need capacity curves, operating rules for modeling
Source: ADASA (Agência Reguladora de Águas, Energia e Saneamento Básico do DF)
Information Needed:
- Maximum storage capacity (m^3)
- Level-volume curve (cota-volume relationship)
- Level-surface area curve (for evaporation calculations)
- Operating rules (target levels, release policies)
- Historical emergency thresholds
Timeline: Week 2-3

6. Calculate or Obtain Evaporation Estimates

Why: Significant outflow component from reservoirs
Options:
- **Option A**: Calculate using Penman-Monteith equation (INMET data)
- **Option B**: Request pan evaporation data from INMET
- **Option C**: Use regional ET coefficients from agricultural sources
Timeline: Week 2-3

7. Request Daily Water Production Data from CAESB

Why: Actual outflow from reservoirs to distribution
Contact: Companhia de Saneamento Ambiental do Distrito Federal
Information Needed:
- Daily production by treatment plant (2014-present)
- Link each plant to source reservoir (Santa Maria or Descoberto)
- Distribution system losses (daily or monthly)
Timeline: Week 2-4 (depends on CAESB response time)

8. Extract SINISA DF-Specific Data

Why: Establish baseline consumption rates and system characteristics
Action:
- Parse Excel sheets for DF (Sigla = "DF")
- Extract all relevant indicators from all categories
- Document indicator codes and definitions
- Calculate derived metrics (e.g., total daily consumption estimate)
Timeline: Week 2

7.3 Medium-Term Actions (Priority 3) 🟡

9. Expand Spatial Coverage (Multi-Station Analysis)

Why: Improve rainfall prediction; capture watershed-scale patterns
Action:
- Identify INMET stations within/near watershed of each reservoir
- Weight stations by proximity and elevation
- Calculate watershed-average precipitation
- Analyze spatial patterns and system movement
Timeline: Week 3-4

10. Obtain Climate Indices

Why: Large-scale patterns influence regional rainfall
Indices to Collect:
- **ENSO (El Niño/La Niña)**: ONI (Oceanic Niño Index) from NOAA
- **South American Monsoon**: Onset and retreat dates
- **Atlantic SST**: Sea surface temperatures (NOAA)
- **MJO (Madden-Julian Oscillation)**: Tropical convection indicator
Source: NOAA, CPTEC/INPE
Timeline: Week 3-4

11. Analyze Regulatory Documents in LAI PDF

Why: Understand historical crisis management and thresholds
Action:
- Extract capacity percentages from pages 10,000+
- Map resolution numbers to dates
- Create timeline of restrictions and interventions
- Identify critical levels for each reservoir
Timeline: Week 4-5

12. Develop Data Dictionary for Modeling

Why: Document final feature set for ML models
Content:
- Feature definitions and units
- Transformation applied (scaling, encoding)
- Missing value treatment
- Expected ranges and outlier thresholds
- Feature importance rankings (from exploratory analysis)
Timeline: Week 4-5

7.4 Data Collection Strategy

Prioritization Framework

```
┌─────────────────────────────────────────────────────────────┐
│  Phase 1: Essential Data (Weeks 1-2)                        │
│  ✓ Descoberto levels ✓ INMET 2019-2024 ✓ Clean datasets    │
└─────────────────────────────────────────────────────────────┘
                            ↓
┌─────────────────────────────────────────────────────────────┐
│  Phase 2: Model-Ready Dataset (Weeks 2-3)                   │
│  ✓ Reservoir specs ✓ Evaporation ✓ SINISA extraction        │
└─────────────────────────────────────────────────────────────┘
                            ↓
┌─────────────────────────────────────────────────────────────┐
│  Phase 3: Enhanced Features (Weeks 3-4)                     │
│  ✓ Multi-station ✓ Climate indices ✓ Consumption data       │
└─────────────────────────────────────────────────────────────┘
                            ↓
┌─────────────────────────────────────────────────────────────┐
│  Phase 4: Context & Validation (Weeks 4-5)                  │
│  ✓ Regulatory history ✓ Documentation ✓ QA complete         │
└─────────────────────────────────────────────────────────────┘
```

7.5 Recommended Model Architectures

Rainfall Prediction Model

Suitable Algorithms (in order of recommendation):

1. **LSTM (Long Short-Term Memory)**
   - **Why**: Excellent for sequence prediction; captures temporal dependencies
   - **Input**: 24-48 hour window of weather variables
   - **Output**: Next hour, next 6 hours, next 24 hours rainfall probability & amount
   - **Advantage**: Handles long-range dependencies, non-linear relationships

2. **XGBoost / LightGBM**
   - **Why**: Powerful with tabular data; fast training; interpretable
   - **Input**: Engineered features (lags, rolling stats, trends)
   - **Output**: Rainfall classification (yes/no) + amount regression
   - **Advantage**: Feature importance analysis, handles missing data well

3. **Ensemble: XGBoost + LSTM**
   - **Why**: Combine strengths of both
   - **Architecture**: LSTM for temporal patterns, XGBoost for feature interactions, ensemble predictions
   - **Advantage**: Highest accuracy potential

Evaluation Metrics:
- **Classification**: ROC-AUC, Precision, Recall (for “will it rain?”)
- **Regression**: RMSE, MAE (for “how much rain?”)
- **Specialized**: CSI (Critical Success Index) for meteorology

Reservoir Level Prediction Model

Suitable Algorithms (in order of recommendation):

1. **Physical Model + ML Correction (Hybrid)**
   - **Physical Component**: Water balance equation
   `Level_t+1 = Level_t + Inflow - Outflow - Evaporation`
   - **ML Component**: LSTM or XGBoost to predict residuals (model errors)
   - **Why**: Combines domain knowledge with data-driven flexibility
   - **Advantage**: Interpretable, respects physical constraints

2. **LSTM (Pure Data-Driven)**
   - **Why**: Captures complex temporal dynamics
   - **Input**: 30-90 day window of reservoir levels, rainfall, consumption, weather
   - **Output**: Next 1, 7, 14, 30 days reservoir level forecast
   - **Advantage**: Can discover non-obvious patterns

3. **XGBoost with Engineered Features**
   - **Why**: Fast iteration, interpretable
   - **Input**: Rich feature set (level lags, rainfall accumulations, seasonal indicators, etc.)
   - **Output**: Daily level predictions
   - **Advantage**: Easy to add new features, feature importance for stakeholder communication

Recommended Approach: **Start with Hybrid Model**
- Use water balance as baseline
- ML model learns correction factors
- Gradually increase ML complexity as data quality improves

Evaluation Metrics:
- **RMSE / MAE**: Absolute level prediction error (cm or m)
- **MAPE**: Percentage error relative to capacity
- **Lead Time Analysis**: Accuracy at 1-day, 7-day, 14-day, 30-day horizons
- **Critical Level Detection**: Did model correctly predict crossing of emergency thresholds?

7.6 SQL Database Schema Recommendation

```sql
-- Core Tables

CREATE TABLE meteorology (
    station_code VARCHAR(10),
    datetime_utc TIMESTAMP,
    datetime_local TIMESTAMP,
    precipitation_mm DECIMAL(10,2),
    temperature_c DECIMAL(5,2),
    dew_point_c DECIMAL(5,2),
    humidity_pct DECIMAL(5,2),
    pressure_mb DECIMAL(7,2),
    wind_speed_ms DECIMAL(5,2),
    wind_direction_deg INTEGER,
    radiation_kjm2 DECIMAL(10,2),
    PRIMARY KEY (station_code, datetime_utc)
);

CREATE TABLE reservoir_levels (
    reservoir_code VARCHAR(20),
    reservoir_name VARCHAR(100),
    datetime TIMESTAMP,
    level_cm INTEGER,
    level_m DECIMAL(10,3),
    capacity_pct DECIMAL(5,2),  -- if capacity curve available
    PRIMARY KEY (reservoir_code, datetime)
);

CREATE TABLE water_consumption (
    date DATE,
    reservoir_code VARCHAR(20),  -- source reservoir
    production_m3 DECIMAL(12,2),
    distributed_m3 DECIMAL(12,2),
    losses_m3 DECIMAL(12,2),
    PRIMARY KEY (date, reservoir_code)
);

CREATE TABLE sinisa_indicators (
    year INTEGER,
    indicator_code VARCHAR(20),
    indicator_name VARCHAR(200),
    value DECIMAL(15,4),
    unit VARCHAR(50),
    PRIMARY KEY (year, indicator_code)
);

CREATE TABLE demographics (
    year INTEGER,
    region VARCHAR(100),
    population INTEGER,
    households INTEGER,
    avg_household_size DECIMAL(4,2),
    -- additional socioeconomic variables
    PRIMARY KEY (year, region)
);

-- Derived/Feature Tables

CREATE TABLE rainfall_features (
    station_code VARCHAR(10),
    date DATE,
    daily_total_mm DECIMAL(10,2),
    max_hourly_mm DECIMAL(10,2),
```

```
    hours_with_rain INTEGER,
    rolling_7d_mm DECIMAL(10,2),
    rolling_30d_mm DECIMAL(10,2),
    days_since_rain INTEGER,
    -- additional engineered features
    PRIMARY KEY (station_code, date)
);

CREATE TABLE reservoir_features (
    reservoir_code VARCHAR(20),
    date DATE,
    avg_level_m DECIMAL(10,3),
    level_change_m DECIMAL(10,3),
    capacity_pct DECIMAL(5,2),
    days_to_critical DECIMAL(10,2),  -- estimated
    watershed_rainfall_mm DECIMAL(10,2),  -- aggregated from stations
    estimated_consumption_m3 DECIMAL(12,2),
    estimated_evaporation_mm DECIMAL(6,2),
    PRIMARY KEY (reservoir_code, date)
);

-- Relationships

-- Meteorology → Reservoir: via watershed aggregation (many-to-one)
-- Reservoir → Consumption: direct link by reservoir_code
-- Demographics → Consumption: via population-based estimation
```

Design Principles:
- **Normalized structure** for raw data tables
- **Denormalized feature tables** for ML pipeline efficiency
- **Datetime consistently in UTC** with local time as separate column
- **All measurements in consistent units** (mm, m, m^3, °C)
- **NULL handling** explicit (use NULL, not -9999 or empty strings)

---

8. FINAL SUMMARY & ACTION PLAN

Data Assets (Current State)

Asset	Status	Quality	Readiness
INMET 2001 Data	✅ Extracted	🟡 Fair (formatting issues)	60%
INMET 2015-2018 Data	✅ Extracted	🟡 Fair (missing radiation)	70%
INMET 2019-2024 Data	⚠️ Available (not processed)	❓ Unknown	0%
Santa Maria Telemetry	⚠️ In PDF	🟢 Good	30% (needs extraction)
Descoberto Telemetry	❌ Not in files	❓ Unknown	0% (need Power BI)
SINISA 2023	✅ Extracted	🟢 Good	40% (needs parsing)
PDAD-DF 2021	✅ Available	🟢 Good	20% (needs extraction)

Overall Readiness: ~40% complete

Critical Path to Model Development

```
Week 1: Data Collection Sprint
├── Extract Descoberto from Power BI ✓
├── Process INMET 2019-2024 ZIPs ✓
├── Extract Santa Maria PDF to CSV ✓
└── Clean INMET data (commas, nulls) ✓

Week 2: Data Integration
├── Create SQL database ✓
├── Import all data sources ✓
├── Extract SINISA DF indicators ✓
├── Calculate evaporation estimates ✓
└── EDA (Exploratory Data Analysis) ✓

Week 3: Feature Engineering
├── Create rainfall features (lags, rolling stats) ✓
├── Create reservoir features (trends, capacity) ✓
├── Engineer interaction features ✓
├── Spatial aggregation (multi-station rainfall) ✓
└── Data validation and QA ✓

Week 4: Model Development - Rainfall
├── Train/val/test split (chronological) ✓
├── Baseline models (persistence, climatology) ✓
├── ML models (XGBoost, LSTM) ✓
├── Hyperparameter tuning ✓
└── Model evaluation and selection ✓

Week 5: Model Development - Reservoir
├── Develop water balance baseline ✓
├── Train ML correction models ✓
├── Multi-horizon prediction (1d, 7d, 14d, 30d) ✓
├── Sensitivity analysis ✓
└── Model evaluation and selection ✓

Week 6: Integration & Deployment Prep
├── Model pipeline integration ✓
├── Create prediction dashboard mockups ✓
├── Documentation (technical + user guide) ✓
└── Stakeholder presentation materials ✓
```

Success Criteria

Rainfall Model:
- ✅ 1-hour ahead: >0.8 ROC-AUC for rain/no-rain classification
- ✅ 24-hour ahead: RMSE < 5mm for rainfall amount
- ✅ 7-day ahead: Capture major rain events (>20mm)

Reservoir Model:
- ✅ 1-day ahead: MAE < 10cm level error
- ✅ 7-day ahead: MAE < 30cm level error
- ✅ 14-day ahead: MAE < 50cm level error
- ✅ 30-day ahead: Correctly identify trend (rising/falling/stable)
- ✅ **Critical**: Correctly predict crossing of emergency thresholds (30% capacity) with 14-day lead time

---

Key Contacts & Resources

Entity	Role	Information Needed
ADASA	Water resources regulator	Reservoir specs, operating rules, inflow data
CAESB	Water utility	Daily production, consumption data by source
INMET	Meteorology agency	Additional station data, pan evaporation
CODEPLAN	DF statistics	Demographic updates, regional breakdowns
ANA	National water agency	Additional hydrological data

APPENDIX

A. Glossary of Terms

Term	Definition
Cota	Water level (elevation) in meters or centimeters
Hidrometração	Water metering (installation of meters on connections)
Macromedição	Bulk water metering (at production/distribution entry points)
Micromedição	Consumer-level water metering (at household/business level)
Perdas de faturamento	Billing losses (water delivered but not billed)
Perdas totais	Total water losses (physical losses + commercial losses)
Runoff coefficient	Fraction of rainfall that becomes surface runoff (0-1)
Antecedent moisture	Soil water content before a rainfall event
Pan evaporation	Water evaporation measured in a standard pan (proxy for reservoir evaporation)

B. Unit Conversions

```
Volume:
1 m³ = 1,000 liters

Flow:
1 m³/day = 1,000 L/day
1 L/capita/day → m³/day: multiply by population, divide by 1,000

Pressure:
1 mbar (mB) = 100 Pa = 1 hPa

Radiation:
1 kJ/m² = 1,000 J/m²
To convert to W/m² (power): divide by time period in seconds

Level:
Cota in cm → m: divide by 100
Cota in m → % capacity: requires reservoir-specific capacity curve
```

C. Data File Locations

```
/home/ubuntu/
├── Uploads/
│   ├── 2001.zip  [INMET historical]
│   │   └── 2001_extracted/
│   ├── SINISA_AGUA_Planilhas_2023_v2.1.1.zip  [Water supply data]
│   │   └── SINISA_extracted/
│   ├── Dados das Barragens(1).pdf  [LAI telemetry - Santa Maria]
│   ├── PDAD-DF_2021.pdf  [Demographics]
│   ├── Relatorio_DF.pdf  [DF general info]
│   └── image.png  [ML Canvas diagram]
│
├── inmet_historico/  [Recent INMET data - NOT YET PROCESSED]
│   ├── 2015.zip
│   ├── 2016.zip
│   ├── ... (2017-2024)
│
├── inmet_A046_GAMA_PONTE_ALTA_2015-2024.csv  [Processed INMET for Gama station]
└── sinisa_indicadores_estruturais_e_operacionais_df.csv  [SINISA DF summary]
```

D. Contact Information for Data Requests

ADASA - Agência Reguladora de Águas, Energia e Saneamento Básico do DF
Website: https://www.adasa.df.gov.br/
Phone: +55 (61) 3961-2200
Request: Reservoir technical specs, historical operational data

CAESB - Companhia de Saneamento Ambiental do Distrito Federal
Website: https://www.caesb.df.gov.br/
Phone: 115
Request: Daily water production data by treatment plant and source reservoir

INMET - Instituto Nacional de Meteorologia
Website: https://portal.inmet.gov.br/
Email: inmet.atendimento@inmet.gov.br
Request: Pan evaporation data, additional station data

ANA - Agência Nacional de Águas e Saneamento Básico
Website: https://www.gov.br/ana/
Request: Hydrological data, streamflow measurements

DOCUMENT VERSION HISTORY

Version	Date	Changes	Author
1.0	2025-10-19	Initial comprehensive data dictionary created	Data Analysis Team

END OF DATA DICTIONARY